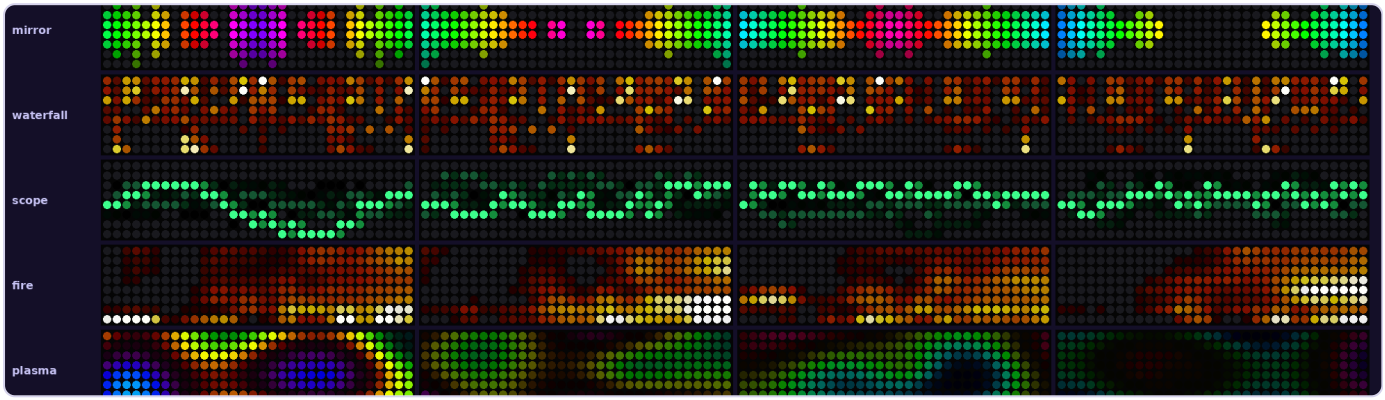
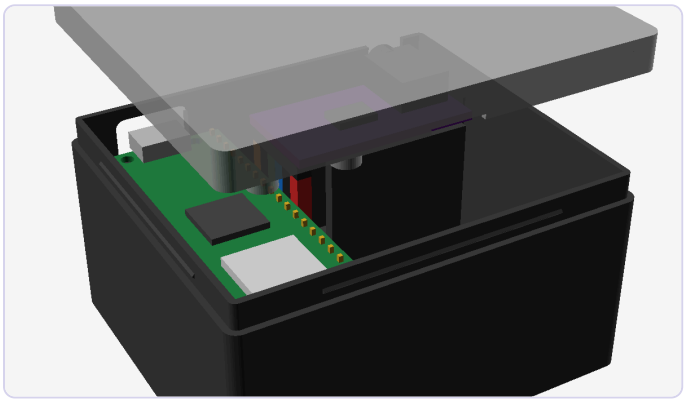
C64 Ecosystem

Running the visualizer, the audio receiver and the Raspberry Pi jukebox together – setup, chaining, operation and special features



Visualizer: light show on an 8 × 32 LED matrix



Audio receiver for the stereo



Pi jukebox in a “breadbin” with a 2.8″ display

1 · Overview
2 · The essentials first
3 · Setting up each device
4 · Chaining the devices
5 · Operation
6 · Special features
7 · Troubleshooting
8 · Notes and license

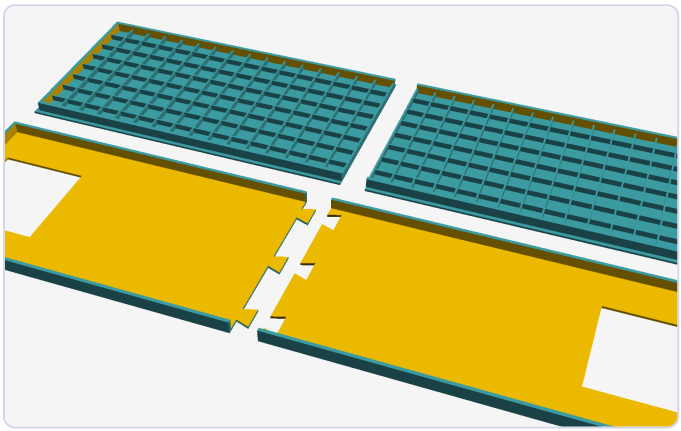
This guide sums up how the three projects work together. Each device also has a detailed manual (in German) – in its web page under the tab **Hilfe** (help) and as `docs/HANDBUCH.md` in its project. **The web pages are in German**; buttons and tabs are quoted here as they appear, in *italics* with a translation. The pictures of the web pages show sample data.

1 · Overview

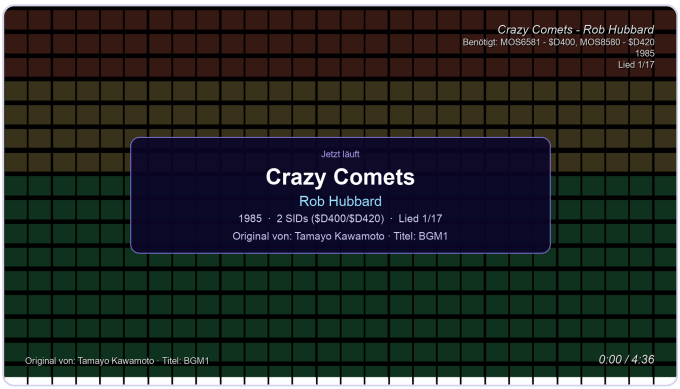
An **Ultimate 64** (Elite, Elite II) or a **C64 Ultimate** can send the C64's sound into the network as an **audio stream** – about 250 packets per second, stereo, 48 kHz. Three home-built devices use this stream:

Device	What it does	Reachable at
Visualizer Raspberry Pi Pico 2 W + WS2812B matrix 8 × 32	Turns the sound into a light show (17 modes). Without music it shows scrolling texts, the time, titles of the jukebox and retro news. Sound from the stream or from an analogue input.	`c64viz.local`
Audio receiver Pico 2 W + PCM5102A DAC module	Plays the stream on the stereo – no cable from the C64 to the stereo, with a jitter buffer and clock tracking.	`c64audio.local`
Pi jukebox Raspberry Pi 3B (or newer)	SID jukebox for the Ultimate (files from a NAS, FTP, USB stick), operated from the phone, picture on the TV – and **stream distributor**: fetches the stream over the network cable and passes it on to all Picos.	`c64pi.local`

None of the devices is required: a visualizer runs on its own with the Ultimate, and so does an audio receiver. Only when several devices should get the same sound does the chaining matter.



Case parts of the visualizer (3D printing, files in the folder `case/` of each project)



The Pi jukebox on the TV: a style with the “now playing” card

The hardware (schematics, parts list, case) is described in the `README.md` of the three projects. Ready-made firmware files are in each project's folder `firmware/`.

2 · The essentials first

> **The Ultimate always streams to one single target only.** When a second device asks for the stream, it takes it away from the first – both then take turns. If several devices should get the sound, **one** of them fetches it and passes it on: either the Pi to all (variant A) or each device to the next one (variant B).

Rule	Why
Always enter the Ultimate's **Ethernet address**	The Ultimates send the stream over the network cable only. The network search of the devices picks the right address automatically.
Unicast instead of multicast	Multicast slows down many WLAN routers: web page and jukebox of the Picos can then hardly be reached. Unicast and forwarding run smoothly (250 packets/s per link, ping ~2 ms).
Picos in the **2.4 GHz WLAN**	The Pico 2 W does not support 5 GHz.
Give every device **its own name**	Otherwise two visualizers are both called `c64viz` (the second one switches to `c64viz-2`). With fixed names like `c64viz2` forwarding finds its target reliably.
Change the start password `commodore64`	Applies to all three web pages and to the devices' own WLANs. On the Picos the browser also asks for a **user name** – any name will do.

The three ways a Pico receives the stream

Setting on the Pico	When
nur dieser Pico (Unicast) only this Pico	The Pico fetches the stream from the Ultimate itself. For the **first** device of a chain or a single device.
von einem anderen Pico weitergeleitet forwarded from another Pico; on the receiver: tick *Stream kommt weitergeleitet*	The Pico **never** asks the Ultimate itself but waits for packets from the Pi or from the device before it. For all devices behind the first one – and for all Picos the Pi supplies.
mehrere Geräte gleichzeitig (Multicast)	Only as an exception, see above.

3 · Setting up each device

Visualizer and audio receiver (Pico 2 W)

1. **Load the firmware:** hold the BOOTSEL button on the Pico, plug in USB, copy the matching `.uf2` from `firmware/` onto the drive that appears (visualizer: `c64viz_wifi_ws2812_usb-350mA.uf2`, or `…_psu-1500mA.uf2` with a power supply; receiver: `c64audio.uf2`). A visualizer that is built in can use its mode button instead of BOOTSEL.
2. **The Pico's own WLAN:** without settings it opens an access point – `C64-Visualizer` or `C64-Audio`, password `commodore64`. Connect your phone and open **http://192.168.4.1** (no password asked here).
3. **Enter your home WLAN** (visualizer: tab *WLAN & Ultimate*, receiver: tab *Einstellungen* / settings): WLAN name and password, a **device name**, your own web password and the Ultimate (*Netzwerk nach Ultimates durchsuchen* – search the network – finds it). Press *Speichern und neu starten* (save and restart).
4. The Pico connects and can then be reached at `http://<device name>.local`. The visualizer's matrix scrolls its address and name, e.g. “WLAN 192.168.1.231 c64viz.local”.

Pi jukebox

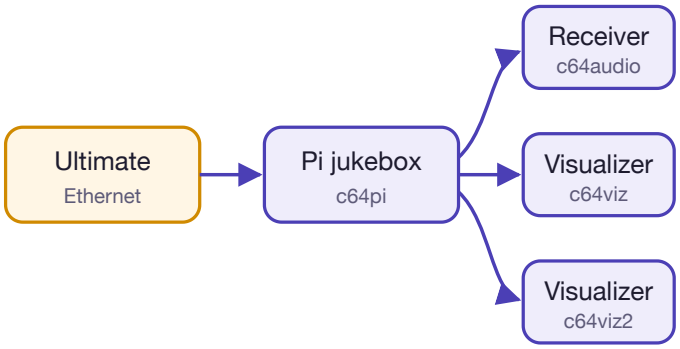
1. Use the **Raspberry Pi Imager** to write “Raspberry Pi OS Lite (64-bit)” to a microSD card; in its settings choose the host name `c64pi`, a user and password, and switch SSH on.
2. Connect the Pi with a **network cable** (the distributor works most reliably over the cable).
3. Over SSH on the Pi (`ssh <user>@c64pi.local`) fetch the project and install it: `sudo apt install -y git`, `git clone https://github.com/sedorrr/c64-pi-jukebox`, `cd c64-pi-jukebox`, `sudo ./install/install.sh`. Or from a Mac/Linux PC in the project folder: `./install/deploy.sh <user>@c64pi.local`.
4. Open **http://c64pi.local/**, log in with `commodore64`, change the password under settings. In the tab *Geräte* (devices) enter the Ultimate (*Im Netz suchen* – search the network) and save.

> Without a cable and without a known WLAN the Pi opens its own WLAN too (`C64-Jukebox`, password `commodore64`, page usually at `http://192.168.5.1/`) – it avoids the Picos' addresses.

4 · Chaining the devices

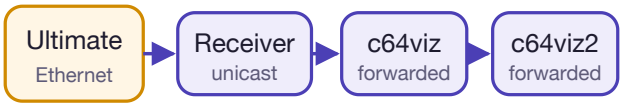
There are two ways to bring the same sound to several devices. Both run smoothly; choose **one**.

A · with the Pi as distributor



cable to the Pi · WLAN to the Picos

B · chain without a Pi

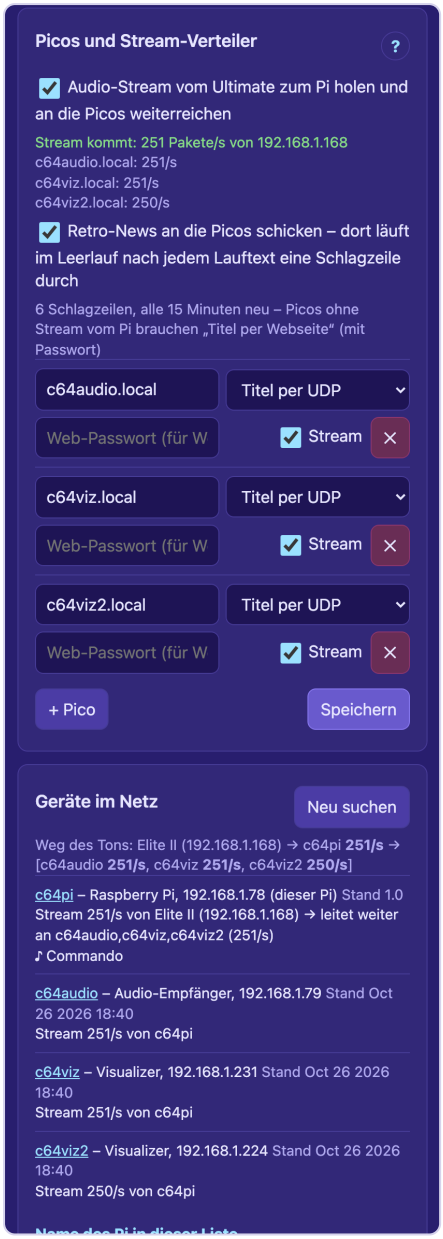


each device passes every packet on to the next one, unchanged

set up with the setup assistant

	A · Pi as distributor	**B · chain without a Pi**
Who fetches the stream?	the Pi (over the cable)	the first device of the chain
If a device fails …	… the others keep their sound; only the Pi is central	… the devices *behind* it get nothing
Titles and retro news	from the Pi to all Picos	titles of the PC jukebox travel along the chain
Setup	list of Picos in the Pi, every Pico on “forwarded”	setup assistant of a visualizer

Variant A: the Pi supplies everyone



Pi, tab *Geräte* (devices): distributor on, three Picos, below them the way of the sound

On the Pi (tab *Geräte*):

1. Under *Ultimates* enter the Ultimate with its cable address and mark it *aktiv* (active).
2. Under *Picos und Stream-Verteiler* (Picos and stream distributor) enter all Picos (`c64audio.local` , `c64viz.local` …) – or tap *als Pico eintragen* (add as Pico) at the bottom under “devices in the network”. For each Pico: *Titel per UDP* (titles over UDP) and tick *Stream*. Save.
3. Tick *Audio-Stream vom Ultimate zum Pi holen und an die Picos weiterreichen* (fetch the stream and pass it on to the Picos). If wanted, also *Retro-News an die Picos schicken* (send retro news to the Picos).

On every visualizer (tab *WLAN & Ultimate*): *Audio-Stream vom Ultimate verwenden* (use the stream) on, receiver mode ***von einem anderen Pico weitergeleitet*** (forwarded), *Stream weiterleiten an* (forward to) **empty**. Save and restart.

On the audio receiver (tab *Einstellungen*): tick ***Stream kommt weitergeleitet*** (stream is forwarded) and switch off forwarding to the visualizer – the Pi supplies everyone directly.

Check: the box under the tick shows “Stream kommt: 251 Pakete/s” and about 250/s for each Pico. Under “devices in the network” (on the Pi and on every Pico) the way of the sound is shown as a tree:

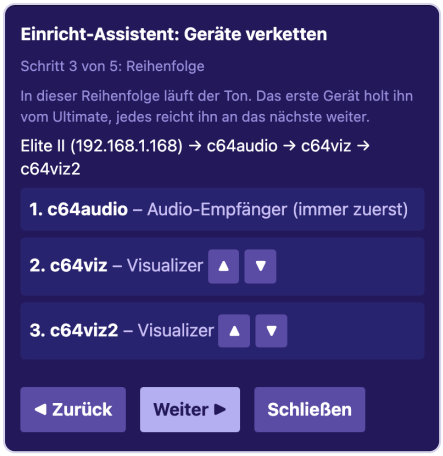
Elite II → c64pi 251/s → [c64audio 251/s, c64viz 251/s, c64viz2 250/s]

> If a Pico is still set to **unicast**, it fetches the stream from the Ultimate itself and thereby takes it away from the Pi. The numbers then flip back and forth between the devices.

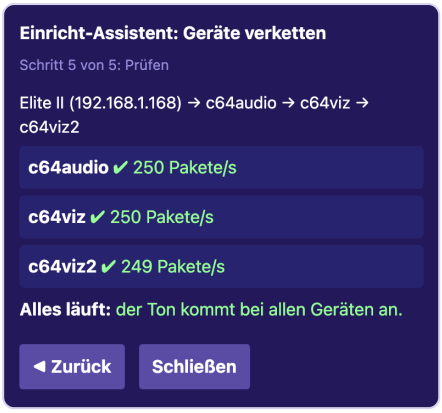
Mixed set-ups work too: the Pi supplies only the audio receiver, which passes it on to the visualizer, and so on. A Pico that would get the stream from two sides takes the first sender and drops the second.

Variant B: chain without a Pi – with the setup assistant

Every visualizer has a **setup assistant** (tab *Steuerung* / control, card “devices in the network”, button *Einricht-Assistent: Geräte verketten*). It finds all Picos in the network and sets them up one after the other:



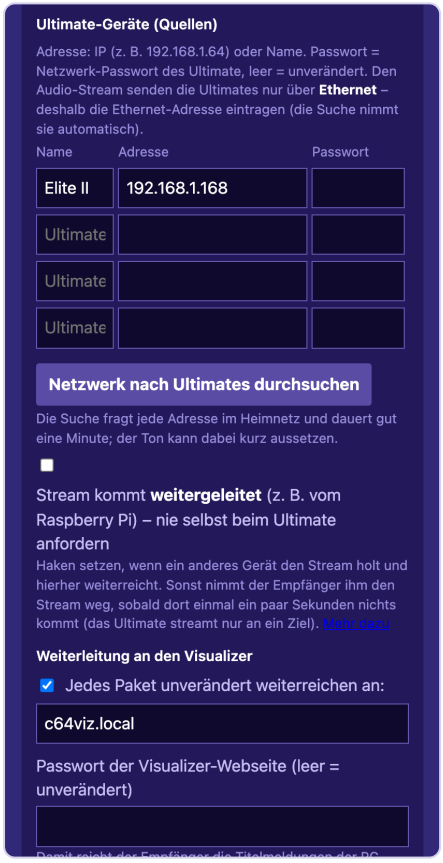
Step 3: order – an audio receiver is always first



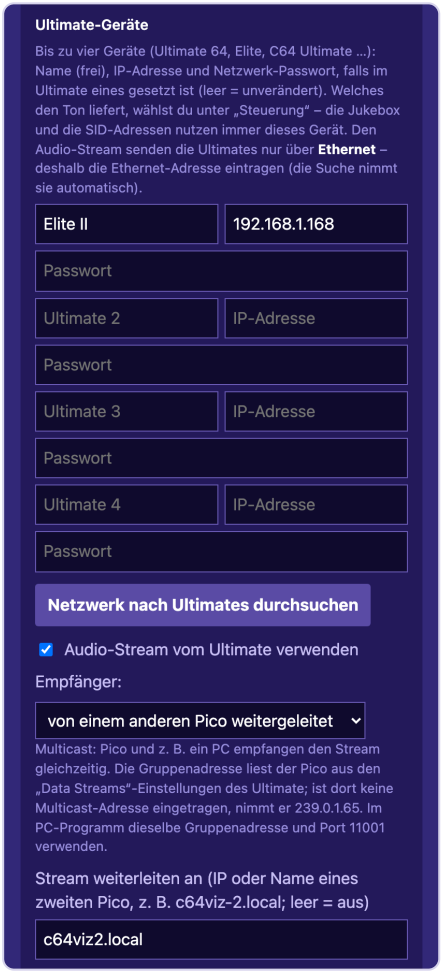
Step 5: check – the sound arrives everywhere

1. **Devices:** tick the ones taking part.
2. **Ultimate:** choose it.
3. **Order:** arrange it (▲ ▼).
4. **Set up:** *Bei … einstellen* (set up at …) opens each device's page with the values already filled in (marked yellow). Log in there with the password of *that* device, check, save. A tick appears as soon as the device reports the new values.
5. **Check:** packets per second at every device, from the right predecessor.

What the assistant sets can also be entered by hand:



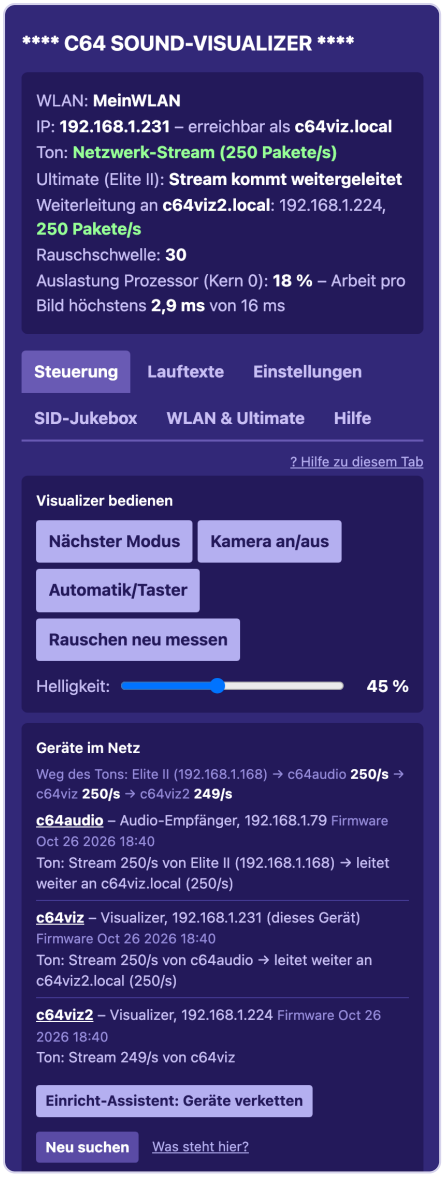
Receiver as first link: fetches itself, passes on to `c64viz.local`



Visualizer in the middle: forwarded, passes on to `c64viz2.local`

Device	Setting
first link	stream on, **unicast** (receiver: tick “forwarded” **off**), forward to the next one
every further one	stream on, **“forwarded from another Pico”**, forward to the next one
last one	as before, forwarding **empty**
all	enter the same Ultimate (for jukebox and SID addresses)

For titles from the PC jukebox to arrive through the audio receiver as well, enter the first visualizer's web password there.



Visualizer, tab *Steuerung*: status box and “devices in the network” with the way of the sound

Check: “devices in the network”

Every Pico and the Pi announce themselves in the home network every 10 s. The card *Geräte im Netz* (devices in the network) shows the same overview on every device:

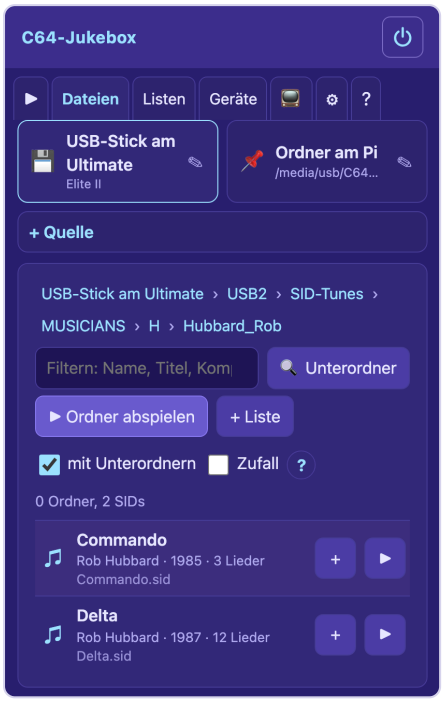
- **Way of the sound**, e.g. “Elite II → c64audio 250/s → c64viz 250/s → c64viz2 249/s”. If a forwarding target gets nothing, it is shown in red at the end.
- For each device: name (a click opens its page), address, firmware version, where it gets the stream from and where it passes it on to.
- **Warnings**, e.g. forwarding that does not run because the stream is off on that device, or a name that cannot be found in the network.

About 250 packets per second at every device means: all is well. Clearly fewer points to multicast or a weak WLAN.

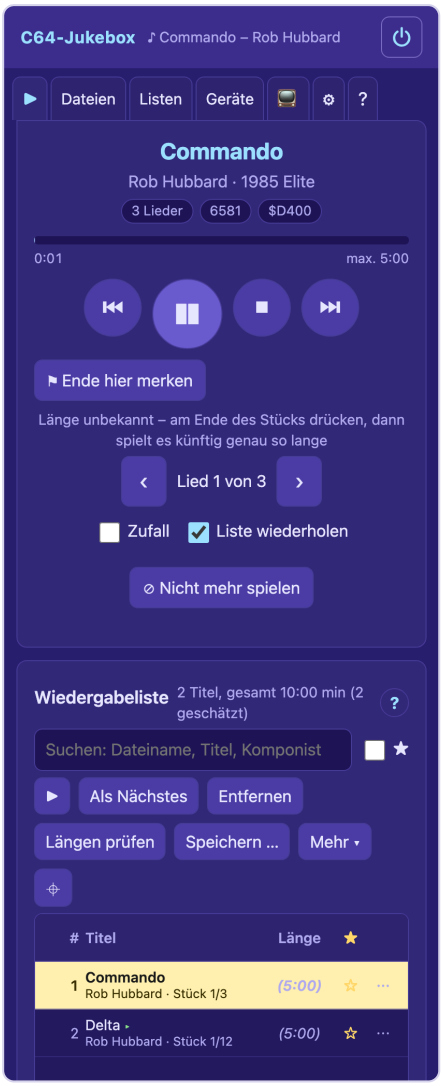
With variant A the way branches at the Pi, e.g. “Elite II → c64pi 251/s → [c64audio 251/s, c64viz 251/s, c64viz2 250/s]” – the same on every device.

5 · Operation

Pi jukebox: choosing and playing music

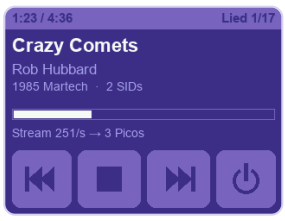


Tab *Dateien* (files): browse folders, filter, tap to play



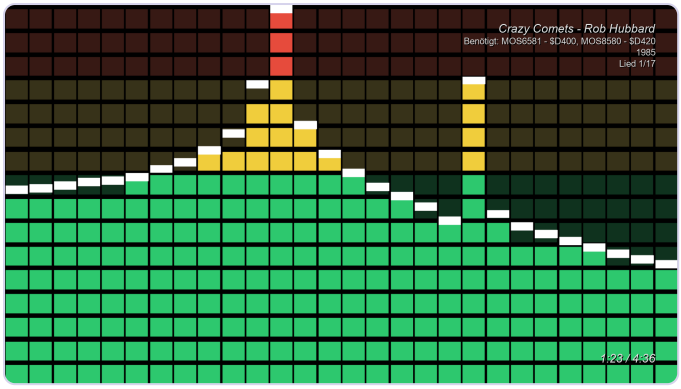
Tab ▶ (playback): current tune, sub-tune, playlist

- **Sources:** network share (NAS, Windows, Mac), FTP server, USB stick on the Ultimate or on the Pi.
- **Playing:** tap a tune; *Ordner abspielen* (play folder) includes all subfolders. Shuffle, repeat list, save playlists.
- **Tune change** exactly at the end with the HVSC song-length file, otherwise after a time limit or silence.
- **Buttons:** ⏮ start/previous tune · ▶/⏸ pause (the Ultimate is paused) · ■ stop (the Ultimate is reset) · ⏭ next tune.
- The jukebox keeps playing on its own, even without an open page.
- On the phone: “Add to Home Screen” – the page then opens like an app.
- **On / Off:** the ⏻ button at the top of every page, “Switch off” in the TV menu, ⏻ on the touch display – the TV, the Ultimate and the jukebox itself; on a Tuya smart plug the power goes off afterwards.
- **Language:** German or English (Settings → System, or “Sprache / Language” in the TV menu).

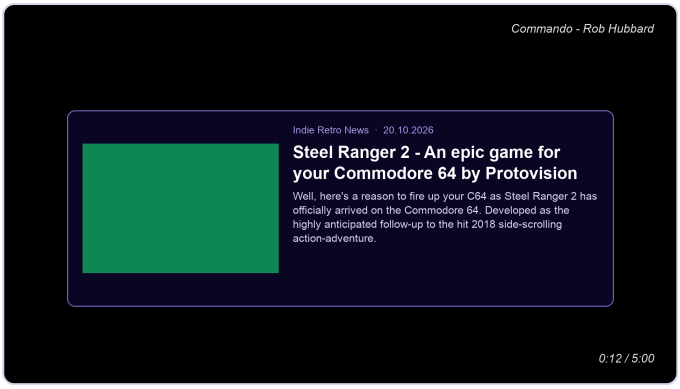


Optional 2.8″ display on the Pi

On the **TV** (HDMI) the Pi shows styles to the music, the SID voices, a slide show or the C64's picture, with title and playing time; it can also be operated with the TV's remote control (HDMI-CEC), and the sound can go to Bluetooth headphones. Settings in the tab 📺 .

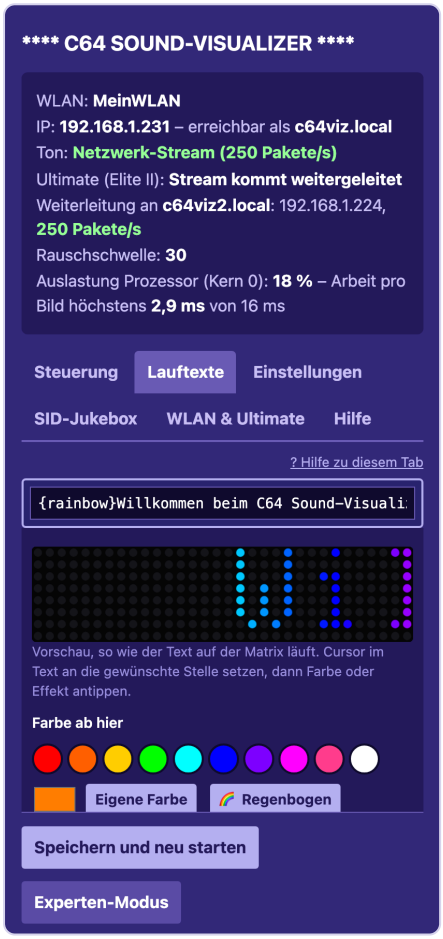


Style “bars” on the TV



Retro news as a card on the TV

Visualizer



Tab *Lauftexte* (scrolling texts) with a preview, just as the text runs on the matrix

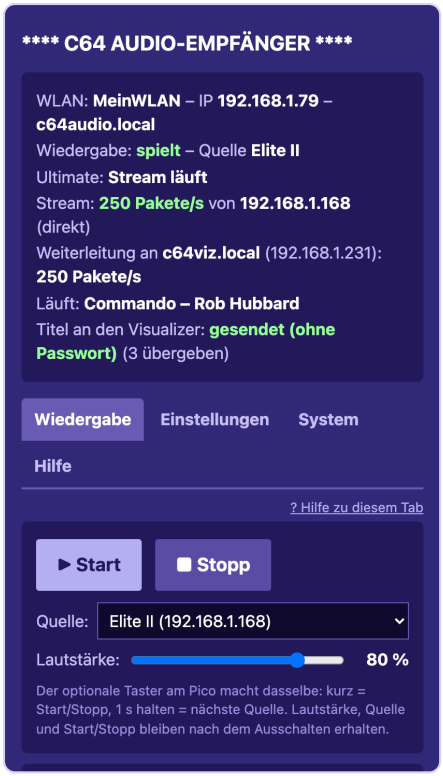
Music starts the light show by itself; after a few seconds of silence the scrolling texts run, with the clock if wanted. It is operated with the **button** or on the web page (tab *Steuerung*, same functions):

Button	Effect
short press	next mode
two short presses	camera mode on/off
three short presses	measure the noise again (stop the music first)
hold 0.7–3 s	brightness 7 → 20 → 45 → 100 %
hold longer than 3 s	automatic mode change or button only
hold while plugging in	USB drive for loading a firmware

Scrolling texts get colours and effects (rainbow, glow, explosion, C64 cursor …) at the push of a button; *Speichern* (save) is all it takes. Further settings (silence time, sleep, night hours, stereo, two matrices, light sensor) are in the tab *Einstellungen*.

The visualizer's own **SID jukebox** plays files from the Ultimate's USB stick or from the PC even without a Pi. When the Pi jukebox is running, the visualizer shows its titles.

Audio receiver



Tab *Wiedergabe* (playback): start/stop, source, volume

- **Start / stop** and **volume** on the page – or with the optional button: short = start/stop, hold 1 s = next source. Volume, source and start/stop are kept after switching off.
- The receiver buffers about 80 ms and follows the Ultimate's clock (about 47,983 Hz); a missing packet is bridged instead of clicking.
- If there are still dropouts (counter *Leergelaufen* – ran empty), a larger buffer under *Einstellungen* helps.
- Turn the volume of the stereo down before the first sound.

6 · Special features

Titles and retro news on the matrix

- For every tune the Pi jukebox sends name, composer, length, sub-tune and SID addresses to the Picos. The matrix shows the title, a faint line at the bottom the progress; a tune with 2 SIDs switches to stereo.
- **Titles over UDP** (no password) reach only Picos that get their stream from the Pi. For all others choose *Titel per Webseite* (titles via web page) in the Pi and enter the Pico's web password.
- **Retro news:** every 30 minutes the Pi fetches headlines from public RSS feeds (Indie Retro News, amiga-news.de, Commodore News, Cascade64, The Oasis BBS …). In idle mode the visualizers show one in yellow after each scrolling text, e.g. “+++ New demo +++”. To switch off: untick it on the Pi or deselect single feeds; on a single visualizer use `@news off` (setting “Retro-News”).

When a device is missing

- If the stream stays away, the visualizer switches to its **analogue input** after 0.25 s (if connected).
- A Pico on “forwarded” never asks the Ultimate itself. If the Pi (variant A) or the device before it (variant B) is off, it stays without a stream.
- Exception in variant B: if the first visualizer behind the receiver is set to **unicast**, it fetches the sound itself while the receiver is off – and gives it back as soon as the receiver runs again.
- If another program (e.g. on the PC) starts a stream, it takes it away from the Pi or the first Pico; they get it back as soon as the program stops.

Several Ultimates

All devices know up to four Ultimates (the Pi up to eight) and one **source**. On the visualizer it is changed in the tab *Steuerung* (not while its jukebox is playing), on the receiver under *Wiedergabe*, on the Pi under *Geräte* (active). With variant A it is enough to change the source on the Pi. An Ultimate with a network password does not answer the network search – then enter it by hand.

Two matrices as one picture

Two visualizers side by side can show one picture of 64 × 8 dots (tab *Einstellungen → Zwei Matrizen als ein Bild*). One of them computes the whole picture and sends its half to the partner – best the one that gets the stream anyway.

Without a router: the Pi's party mode

Away from home the Pi opens a WLAN itself – with the name and password of your home WLAN. Picos and phones then connect as at home; the Ultimate is connected to the Pi directly by cable (settings → *Party-Modus*).

Updates

- **Picos:** tab *WLAN & Ultimate* (receiver: *System*) → *Firmware (.uf2) wählen …* (choose firmware). All Picos at once: `python3 tools/update_all.py` in the visualizer project. Settings and texts are kept.
- **Pi:** the page and the TV menu show a new version on GitHub; “Install” (or “Check for update” in the TV menu) downloads and installs it. From version 1.0 once by hand: get the new project folder and run `install.sh` or `deploy.sh` again. Settings and lists are kept.
- **Backup:** every device can download all its settings as one file and restore them.

7 · Troubleshooting

What you see	Cause and remedy
Two devices get sound alternately	Both fetch the stream themselves. Set all devices except the first one (or all Picos with variant A) to **“forwarded”**.
wartet auf weitergeleiteten Stream (waiting for forwarded stream)	The device before it has no forwarding, or its stream is off – or (variant A) the Pico is not in the Pi's list or its “Stream” tick is missing.
Name wird gesucht …, *Name nicht gefunden* (name not found)	Device off, different name or in another WLAN. If need be, enter the IP address instead of the name.
Clearly fewer than 250 packets/s, sluggish web page	Multicast is still set somewhere, or the WLAN is weak. Choose unicast or “forwarded” everywhere.
No stream although everything is entered	The Ultimate's WLAN address was entered instead of its Ethernet address; or the Ultimate's web remote control (Network → Web Remote Control) is off.
Pi: “… Pakete von <address> verworfen” (packets dropped)	The Ultimate sends from an address that is not entered in the Pi (usually its WLAN address). Enter the cable address.
Titles do not appear on the matrix	The Pico does not get the stream from the Pi → choose “titles via web page” with the Pico's password in the Pi.
A Pico ends up in its own WLAN (C64-Visualizer / C64-Audio)	Home WLAN not found or wrong password. It tries again every 5 minutes; correct the data at http://192.168.4.1.
Password forgotten	Picos: load `wlan_reset.uf2` (clears only WLAN, Ultimate and password). Pi: reset it over SSH, see the Pi manual.

Every device has a **diagnostics log** (visualizer: *WLAN & Ultimate → Diagnose laden*, receiver: *System*). The detailed troubleshooting chapters are in the manuals (tab *Hilfe*).

8 · Notes and license

- **Home network only.** The web pages use plain HTTP without encryption. Stream, titles, retro news, device messages and the picture for two matrices travel over UDP **without any login** – every device in the same network can send them. Do not make the devices reachable from the internet (no port forwarding) and do not run them in foreign or public networks.
- **At your own risk.** Work on the C64 or the Ultimate, building the devices and the power supply of the LED matrix (several amperes at full brightness) are at your own risk. No liability is accepted for damage to devices, hardware or data.
- **Retro news** show only headlines with their source; the content belongs to the respective pages.
- Only tested with the author's hardware and home network (Pico 2 W, WS2812B 8 × 32, PCM5102A, Raspberry Pi 3B, Ultimate 64 Elite II). Other routers and Ultimate firmwares may behave differently.

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